

2(a)(i)	$\Delta p = mv - mu$ $= 0.14(-4.0 - (5.4))$ $= -1.3 \text{ N s}$	
2(a)(ii)	$F = \left \frac{\Delta p}{\Delta t} \right = \frac{1.3}{0.04} = 32.5 \text{ N}$ $F = N - mg$ $32.5 = N - 0.14(9.81)$ $N = 34 \text{ N}$ By Newton's 3rd law, force exerted by the ball on the bar = 34 N	
2(b)(i)	Taking moment about B, $34(75) + 0.45(9.81)(25) = F_A(20)$ $F_A = 133 \text{ N}$	

2(b)(ii)	$F_B = F_A + 34 + 0.45(9.81)$ $F_B = 171 \text{ N}$	
3(a)	The gas consists of a <i>very large number of molecules in</i>	